Small Engines Overhaul · Unit 8 · Level 2

Engine Teardown & Rebuild



TODAY'S PLAN

- 👥 Jobs
- ⚠️ Safety
- 🔧 Steps
- ✅ Check
- 📝 Reflect

Get ready

Jobs



Mechanic

Does the main engine job with the tools



Tool helper

Gets tools from the board and returns them to staff



Recorder

Takes photos and fills in the chart



Clean-up captain

Wipes the bench, sweeps, checks tools are back

Safety

Be Safe



- 🥽 **Safety glasses** on the whole time.
- 👖 **Coveralls** and closed-toe shoes; tie back long hair.
- 🔧 Use tools **with adult supervision**. Right tool for the job.
- 🌀 Valve and starter **springs** store energy: eyes protected, tool held with adult supervision.
- 🏭 Engines are started **outdoors** with an adult (carbon monoxide).
- 🧼 **Clean up** and wash hands when you finish.

Steps

Steps 1 of 2



1. 🔌 Disconnect plug wire; with adult supervision drain fuel and oil.
2. 📸 Photo; remove shroud, air filter, carb, muffler (tray cups 1-4).
3. 🔘 Remove flywheel with the puller; remove armature.
4. 🔩 Remove head; remove valves with the spring compressor.
5. ⭕ Remove piston/rod; check ring end gap and side clearance; record.

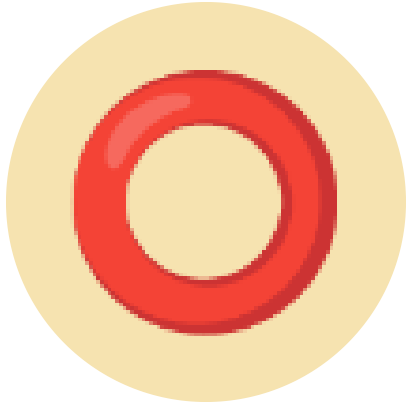
Steps

Steps 2 of 2



1. 🔄 Lap valves; set valve clearance to spec.
2. ⏱️ Reassemble: align timing marks; new gaskets; oil parts.
3. 🔧 Torque head bolts in pattern to spec.
4. 🔍 Test: fuel, spark, compression; adult starts the engine outdoors.

🃏 Check 1 of 3



End gap is measured…

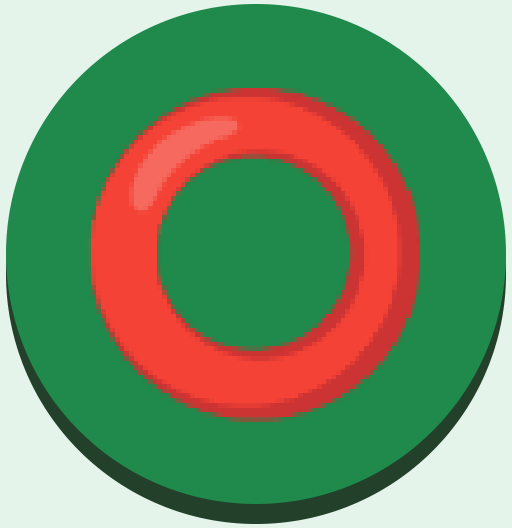
Hold up 1, 2, or 3:

1️⃣ Ring in the bore, feeler gauge

2️⃣ Ring on the floor

3️⃣ With a ruler

✅ Answer · Card 1 of 3



Ring in the bore, feeler gauge

Bore.

🃏 Check 2 of 3



The valve spring compressor…

Hold up 1, 2, or 3:

1️⃣ Makes spark

2️⃣ Mixes fuel

3️⃣ Holds the spring to remove keepers

✅ Answer · Card 2 of 3



Holds the spring to remove keepers

Springs.

🃏 Check 3 of 3



Timing marks must…

Hold up 1, 2, or 3:

1️⃣ Be painted

2️⃣ Line up

3️⃣ Be ignored

✅ Answer · Card 3 of 3



Line up

Valve timing.

Wrap-Up

Clean Up + Reflect



Return and count tools.

Sweep and wipe your bench.

Wash hands, then do the reflection.